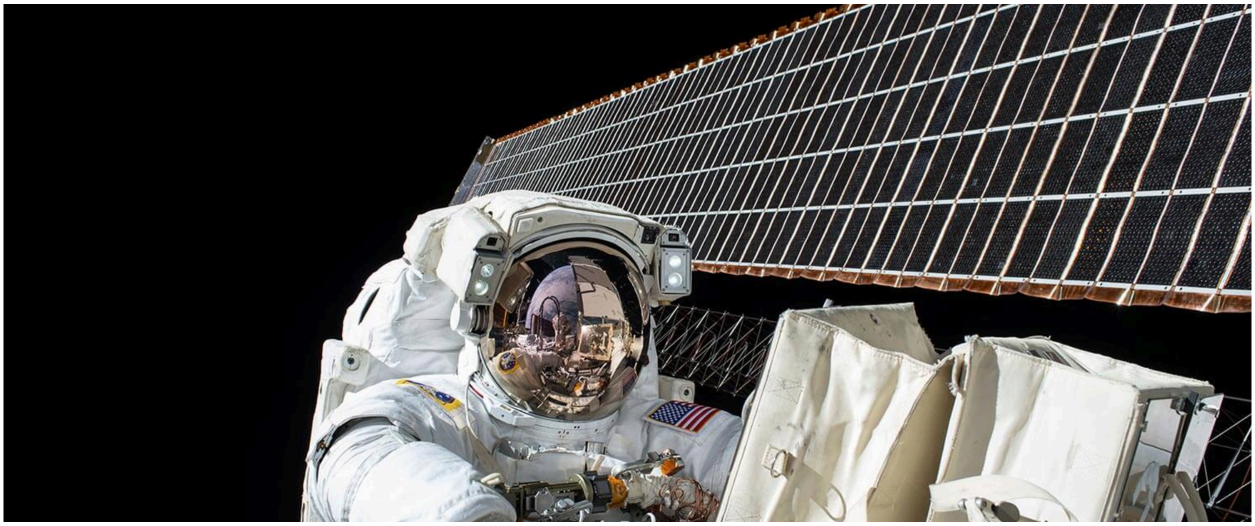

High Availability, Not High Throughput: Classic Active/Passive HAProxy Failover with Keepalived

> Highlights that only one node serves traffic at a time—reliability over performance scaling.

Single Keepalived Setup - Pros and Cons

✅ ADVANTAGES (Why it's WORTH it):

1. **Simpler Architecture** - Less complexity, easier to manage
2. **No WiFi VRRP Issues** - WiFi often blocks VRRP multicast traffic
3. **More Reliable** - No risk of "split-brain" scenarios
4. **Easier Troubleshooting** - Fewer moving parts
5. **Better Performance** - No constant health checks between two LBs

❌ DISADVANTAGES:

1. **Single Point of Failure** - If Keepalived VM goes down, VIP fails
2. **Manual intervention needed** if the primary LB fails

Recommended Architecture:

```
        ┌─────────────────┐
        │   CLIENT        │
        └─────────────────┘
                 ↓
        ┌─────────────────┐
        │ VIP: 192.168.0.10 │ ← Always on VM
        └─────────────────┘
                 ↓
    ┌────────────────────────────────────────┐
    │                 │
┌───────────────┐          ┌───────────────┐
│ 192.168.0.105 │          │ 192.168.0.104 │
│   VM          │          │  Laptop       │
│ Keepalived    │          │   Nginx       │
│ HAProxy       │          │  (Standby)    │
│  ACTIVE       │          │               │
└───────────────┘          └───────────────┘
```

How to Set Up Single Keepalived:

On 192.168.0.105 (VM - Keepalived + HAProxy):

Keep the current Keepalived config:

```
global_defs {
   enable_script_security
}

vrrp_script chk_haproxy {
   script "/usr/bin/killall -0 haproxy"
```

```
    interval 2
    weight 50
}

vrrp_instance VI_WAZUH {
    state MASTER
    interface ens33
    virtual_router_id 51
    priority 150
    advert_int 1
    authentication {
        auth_type PASS
        auth_pass wazuh_secure_789
    }

    track_script {
        chk_haproxy
    }

    virtual_ipaddress {
        192.168.0.10/24
    }
}
```

On 192.168.0.104 (Laptop - Nginx Only):

Stop and disable Keepalived:

```
sudo systemctl stop keepalived
sudo systemctl disable keepalived
sudo systemctl enable nginx
sudo systemctl start nginx
```

Manual Failover Process (If VM Fails):

If 192.168.0.105 goes down, you manually:

1. **Start HAProxy on laptop** (if you install it)
2. **Change client configurations** from `192.168.0.10` to `192.168.0.104`
3. **Or** assign the VIP manually to laptop (advanced)

Even Better: Install HAProxy on Both Machines

You can have both load balancers ready:

- **192.168.0.105**: Keepalived + HAProxy (Active)
- **192.168.0.104**: HAProxy only (Standby)

Then if VM fails, you just:

1. Start HAProxy on laptop
2. Manually assign VIP or change DNS

My Recommendation: ✅ GO FOR IT!

For your setup, **single Keepalived is better** because:

1. **WiFi limitations** make two-node VRRP unreliable
2. **Your VM is more stable** than a laptop
3. **Simpler to manage and troubleshoot**
4. **Good enough for most use cases**

Client Configuration:

All your clients connect to **VIP: 192.168.0.10**

Wazuh Agents:

```
<client>
  <server>
    <address>192.168.0.10</address>
    <port>15140</port>
```

```
  </server>
</client>
```

Dashboard: `https://192.168.0.10:8443`

Monitoring:

Set up simple monitoring to alert you if the VIP becomes unavailable:

```
# Simple ping monitor
ping -c 3 192.168.0.10
```

Bottom Line:

Yes, it's absolutely worth it! Single Keepalived gives you:

- Automatic failover for HAProxy services
- Simple architecture
- Reliable operation (no WiFi VRRP issues)
- Easy maintenance

The only downside is manual intervention if the entire VM fails, but for most environments, this is an acceptable trade-off for the increased reliability and simplicity.

Would you like me to help you clean up the 192.168.0.104 configuration and set up the single Keepalived properly?